

Closed-View Auditing and CPPCs (Addendum): Multi-Stream Monitoring Notes and Schema Excerpts

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Abstract

This companion note collects two items that are useful operationally but intentionally excluded from the brief main CPPC paper (State 3): (i) a short multi-stream monitoring note (e-values and scan inflation), and (ii) a concrete CPPC schema excerpt illustrating what a machine-checkable “control-plane privacy card” can look like.

Scope. Addendum to State 3.[1]

Imports. We import the CPPC concept, the measurement-firewall pattern, and the alert-oracle framing from State 3.[1] Contract objects and publication semantics follow Synthesis 0 and ABOM/OINL.[2, 3] Artifact availability follows the artifact policy.[4]

Exports. A small schema excerpt and a compact “many-detectors” monitoring recipe.

A Multi-stream monitoring note (e-values)

Stateful anonymity deployments often monitor many candidate witnesses (many committees/keys/relays/metrics). A concise way to control “scan inflation” is to attach each detector an *e-process* (a nonnegative supermartingale with unit expectation under the null) and apply e-value multiple testing.

E-values and e-BH. E-values generalize likelihood-ratio betting scores; they compose naturally under optional stopping and enable dependence-robust FDR control via the e-BH procedure [5, 6]. Operationally, this lets an operator run many closed-view detectors while bounding the expected false discovery proportion.

Stopping-time subtlety. When alerts are raised at data-dependent global stopping times (“whenever anything looks bad”), additional care is required; the stopped e-BH procedure is designed for this regime [7]. the main paper (alerts section) develops the deployment-side implication: *publishing* the alert time/window can itself be a witness stream and should be priced.

B CPPC schema (excerpt) and example cards

To keep this note referee-tight, we include only a short *excerpt* of the CPPC schema and two illustrative cards. The full JSON schema and complete card examples (including machine-checking tooling) are available on request as part of the series archive.

B.1 Schema excerpt (v1.1)

```
{
  "schema_version": "1.1",
  "system": "string",
  "unit_of_privacy": { "...": "..." },
  "witness_streams": [ "string", "..." ],
  "witness_channels_present": [
    "semantic_outputs",
    "budget_beacon",
    "transparency_log",
    "timing_and_metadata",
    "auditing_alerts"
  ],
  "cadence": { "epoch": "...", "publication": "..." },
  "view_accountant": {
    "notion": "max-divergence | RDP | f-DP/GDP",
    "composition": "...",
    "budget": { "...": "..." }
  },
  "budget_beacon": { "cadence": "...", "format": "..." },
  "transparency": { "log": "...", "signatures": "..." },
  "compliance": { "mode": "...", "artifacts": [ "..." ] }
}
```

B.2 Reference linter (excerpt)

In the draft archive we include a small reference linter (`tools/cppc_lint.py`) that checks a CPPC JSON for required fields and basic internal consistency. The intent is to make “machine-checkable” concrete without requiring any particular SBOM or transparency-log ecosystem.

```
$ python tools/cppc_lint.py card.json --root ./releases/
```

```
OK: required keys present
```

```
OK: witness_channels_present is a subset of allowed enum
```

```
OK: compliance.artifacts exist under --root and match declared sha256 (if provided)
```

```
WARN: transparency.log missing (deployment-specific)
```

```
WARN: transparency.signatures missing (deployment-specific)
```

B.3 Example CPPC (Nym-style, illustrative excerpt)

```
{
  "schema_version": "1.1",
  "system": "nym-mixnet (illustrative)",
  "unit_of_privacy": { "message": "1 packet-equivalent" },
  "witness_channels_present": [
    "budget_beacon",
    "auditing_alerts",
    "timing_and_metadata"
  ]
}
```

```

],
"cadence": {
  "epoch": "1 min",
  "publication": "daily aggregates; delayed by 7d"
},
"witness_streams": [
  "mix-queue depth (bucketed)",
  "node reputation updates (coarsened)",
  "drift alerts (delayed, bucketed)"
],
"view_accountant": {
  "notion": "RDP",
  "composition": "odometer (adaptive cadence)",
  "budget": { "epsilon_rdp": 3.0, "delta": 1e-6, "horizon": "30d rolling" }
},
"budget_beacon": {
  "cadence": "daily",
  "payload": "current spend + remaining budget (coarsened)"
},
"transparency": {
  "log": "append-only merkle log",
  "signatures": "operator key + optional attestation"
},
"compliance": {
  "mode": "attestation",
  "artifacts": [ "cppc.json", "build hash", "release hashes" ]
}
}

```

B.4 Example CPPC (Tor-style, illustrative excerpt)

```

{
  "schema_version": "1.1",
  "system": "tor-relay-selection (illustrative)",
  "unit_of_privacy": { "client": "one long-lived guard set" },
  "witness_channels_present": [
    "semantic_outputs",
    "budget_beacon",
    "transparency_log"
  ],
  "cadence": {
    "epoch": "1 consensus period",
    "publication": "public consensus + coarse metrics"
  },
  "witness_streams": [
    "selection weights (public consensus)",
    "bandwidth measurement summaries (coarse)",

```

```

    "guard selection randomness"
  ],
  "view_accountant": {
    "notion": "max-divergence",
    "composition": "filter (abort-safe)",
    "budget": { "epsilon": 2.0, "delta": 1e-6, "horizon": "90d" }
  },
  "budget_beacon": {
    "cadence": "each consensus period",
    "payload": "monotone spend counter + checkpoint hash"
  },
  "transparency": {
    "log": "public merkle log (checkpointed)",
    "signatures": "authority keys"
  },
  "compliance": {
    "mode": "hybrid",
    "artifacts": [ "cppc.json", "consensus hashes", "auditor scripts (on request)" ]
  }
}

```

References

- [1] Anonymous. Closed-View Auditing for Stateful Anonymity: Measurement Firewalls and CPPCs. AnonDHT State series Paper 3, 2026. (This archive.)
- [2] Anonymous. Contract Objects for Anonymous-DHT Privacy Budgets and Claims. Series synthesis note (Synthesis 0), 2026. (This archive.)
- [3] Anonymous. ABOM/OINL: Audit BOM and Outcome Identification Noise Label for Anonymous DHT Deployments. Anonymity series Paper C, 2026. (This archive.)
- [4] Anonymous. Artifact policy and supporting materials for the anonymity/AnonDHT series. 2026.
- [5] Aaditya Ramdas and Ruodu Wang. Hypothesis testing with e-values. arXiv:2410.23614, 2024. DOI: 10.48550/arXiv.2410.23614. <https://arxiv.org/abs/2410.23614>
- [6] Ruodu Wang and Aaditya Ramdas. False discovery rate control with e-values. *Journal of the Royal Statistical Society: Series B*, 84(3), 822–852, 2022. DOI: 10.1111/rssb.12489.
- [7] Hongjian Wang, Sanjit Dandapanthula, and Aaditya Ramdas. Anytime-valid FDR control with the stopped e-BH procedure. arXiv:2502.08539, 2025. DOI: 10.48550/arXiv.2502.08539. <https://arxiv.org/abs/2502.08539>